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Object-Oriented Programming — Concepts

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Object-Oriented Programming — Concepts

Course Outline | 16 Weeks | 1 Hour/Week

Institution: IIC University of Technology

Course URL: <http://elearning.iic.edu.kh/course/view.php?id=1229>

Illustration Language: Java

Course Description

This course introduces the fundamental concepts and principles of **Object-Oriented Programming (OOP)**. Students will learn how to think in an object-oriented way — modeling real-world problems as interacting objects — and understand the four core pillars: **Encapsulation, Inheritance, Polymorphism, and Abstraction**. Java is used solely as a vehicle to illustrate and practice these concepts. All topics are kept at the **introductory level**. Advanced OOP topics (SOLID Principles, Design Patterns, Generics, Streams) are covered in the follow-on course.

Course Schedule Overview

Week	Date Range	Topic	Type
1	27 May – 2 June	Introduction to OOP	Lecture
2	3 June – 9 June	Classes & Objects	Lecture + Lab
3	10 June – 16 June	Attributes, Methods & Object State	Lecture + Lab
4	17 June – 23 June	Encapsulation	Lecture + Lab
5	24 June – 30 June	Constructors & Object Initialization	Lecture + Lab
6	1 July – 7 July	Inheritance — IS-A Relationship	Lecture + Lab
7	8 July – 14 July	Method Overriding & the super Reference	Lecture + Lab
8	15 July – 21 July	MID-TERM EXAM	Exam

9	22 July – 28 July	Polymorphism	Lecture + Lab
10	29 July – 4 August	Abstraction — Abstract Classes	Lecture + Lab
11	5 August – 11 August	Abstraction — Interfaces	Lecture + Lab
12	12 August – 18 August	Object Relationships (HAS-A)	Lecture + Lab
13	19 August – 25 August	Static Members & the Object Class	Lecture + Lab
14	26 August – 1 September	Exception Handling Basics	Lecture + Lab
15	2 September – 8 September	Course Review & Mini OOP Project	Review + Lab
16	9 September – 15 September	FINAL EXAM	Exam

Course Architecture

OOP Concepts – Basic Level – 16 Weeks

- PART 1: FOUNDATIONS (Weeks 1–3)
 - Week 1: What is OOP? History, Benefits, OOP vs Procedural
 - Week 2: Classes & Objects – The Core Building Blocks
 - Week 3: Attributes, Methods & Object State
- PART 2: THE FOUR PILLARS (Weeks 4–7)
 - Week 4: Encapsulation
 - Week 5: Constructors & Object Initialization
 - Week 6: Inheritance & IS-A Relationship
 - Week 7: Method Overriding & the super Reference
- ★ MID-TERM EXAM (Week 8) – Covers Weeks 1–7 ★
- PART 3: ABSTRACTION & POLYMORPHISM (Weeks 9–11)
 - Week 9: Polymorphism
 - Week 10: Abstraction – Abstract Classes
 - Week 11: Abstraction – Interfaces
- PART 4: OOP IN PRACTICE – BASICS (Weeks 12–15)
 - Week 12: Object Relationships (HAS-A)
 - Week 13: Static Members & the Object Class
 - Week 14: Exception Handling Basics
 - Week 15: Course Review & Mini OOP Project
- ★ FINAL EXAM (Week 16) – Covers Weeks 1–15 ★

What This Course Covers vs What Comes Next

THIS COURSE (Basic OOP)	NEXT COURSE (Advanced OOP)
Classes, Objects, Constructors	Generics & Type Parameters
Encapsulation	Collections Framework (deep)
Inheritance & IS-A	Lambda Expressions
Method Overriding	Streams API
Polymorphism	SOLID Design Principles
Abstract Classes	Design Patterns (GoF)
Interfaces (basic)	Annotation Processing
Object Relationships (HAS-A)	Reflection API
Static Members	Concurrency Basics
Basic Exception Handling	Advanced Exception Hierarchies

Recommended Reading

Resource	Type	Access
<i>Introduction to Programming Using Java</i> — David J. Eck	Free PDF / HTML	https://math.hws.edu/javanotes/
<i>OOP Using Java</i> — Simon Kendal	Free PDF	https://www.minds.co.za/wp-content/uploads/2019/06/object-oriented-programming-using-java.pdf
<i>Java, Java, Java</i> — Morelli & Walde	Open Textbook	https://open.umn.edu/opentextbooks/textbooks/java-java-java-object-oriented-problem-solving
Oracle Java OOP Tutorial	Official Reference	https://docs.oracle.com/javase/tutorial/java/concepts/

Course Outline — Object-Oriented Programming Concepts (Basic)

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